

# ☒ 40 MCQs on Session 6: Python Basics

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## ☒ Easy Level (10 MCQs)

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Q1. Which command is used to check Python version after installation?

- A. python --check
- B. python --version
- C. check python
- D. version python

**Answer:** B

**Explanation:** python --version shows the installed Python version.

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Q2. What does print("Hello") do?

- A. Prints Hello to console
- B. Declares a variable
- C. Starts a function
- D. Opens a file

**Answer:** A

**Explanation:** print() outputs text.

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**Q3.** How do you declare a function in Python?

- A. function greet():
- B. def greet():
- C. func greet()
- D. declare greet()

**Answer:** B

**Explanation:** def keyword defines a function.

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**Q4.** What symbol indicates a block of code in Python?

- A. {}
- B. ()
- C. Indentation (spaces/tabs)
- D. <>

**Answer:** C

**Explanation:** Indentation defines code blocks in Python.

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**Q5.** Which data type holds text?

- A. int
- B. str
- C. bool
- D. float

**Answer:** B

**Explanation:** str is for strings (text).

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**Q6.** What will `type(10.5)` return?

- A. int
- B. float
- C. str
- D. bool

**Answer:** B

**Explanation:** 10.5 is a floating point number.

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**Q7.** What keyword runs code only when a module is executed directly?

- A. `if __name__ == "__main__":`
- B. `if module == "__main__":`
- C. `run module`
- D. `main()`

**Answer:** A

**Explanation:** This block runs when the file is the main program.

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**Q8.** Which of these is mutable?

- A. tuple
- B. list
- C. str
- D. int

**Answer:** B

**Explanation:** Lists can be changed after creation; tuples cannot.

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**Q9.** Which keyword is used to import a module?

- A. include
- B. import
- C. use
- D. module

**Answer:** B

**Explanation:** import brings in external modules.

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**Q10.** What is an object in Python?

- A. A variable only
- B. An instance of a class
- C. A data type
- D. A function

**Answer:** B

**Explanation:** Objects are instances of classes.

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☒ **Medium Level (20 MCQs)**

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**Q11.** Which of the following is NOT a valid Python function declaration?

- A. `def my_func():`
- B. `def myFunc():`
- C. `function my_func():`
- D. `def _my_func():`

**Answer:** C

**Explanation:** function keyword is not used in Python.

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**Q12.** What will this code print?

```
python
```

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```
def greet(name): print("Hello " + name) greet("Komal")
```

- A. Hello name
- B. Hello Komal
- C. Error
- D. Nothing

**Answer:** B

**Explanation:** Function concatenates and prints the string.

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**Q13.** Which data type is immutable?

- A. list
- B. dict
- C. tuple
- D. set

**Answer:** C

**Explanation:** Tuples cannot be changed after creation.

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**Q14.** What does this code snippet demonstrate?

python

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```
if __name__ == "__main__": print("Testing")
```

- A. Function call
- B. Module test code

- C. Syntax error
- D. Class definition

**Answer:** B

**Explanation:** Code inside runs only if file executed directly.

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**Q15.** Which built-in function converts a string to an integer?

- A. str()
- B. int()
- C. float()
- D. bool()

**Answer:** B

**Explanation:** int() converts strings or numbers to integers.

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**Q16.** What will len("Python") return?

- A. 5
- B. 6
- C. 7
- D. Error

**Answer:** B

**Explanation:** Length of string "Python" is 6 characters.

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**Q17.** What is the output of this code?

python

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```
a = [1, 2, 3] a[0] = 10 print(a)
```

- A. [1, 2, 3]
- B. [10, 2, 3]
- C. Error
- D. [1, 10, 3]

**Answer:** B

**Explanation:** Lists are mutable, so index 0 is updated.

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**Q18.** What happens if indentation is incorrect?

- A. Code runs normally
- B. IndentationError
- C. Warning only
- D. Code compiles differently

**Answer:** B

**Explanation:** Python requires consistent indentation.

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**Q19.** How do you import only the sqrt function from math module?

- A. import math.sqrt
- B. from math import sqrt
- C. import sqrt from math
- D. include math.sqrt

**Answer:** B

**Explanation:** This imports only sqrt from math.

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**Q20.** Which of these is the correct way to define an empty function?

A.

```
python
```

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```
def func(): pass
```

B.

python

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def func(): None

C.

python

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def func(): return

D. All of the above

**Answer:** D

**Explanation:** All are valid to define empty functions.

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## ☒ Hard Level (10 MCQs)

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Q21. What is the type of the expression: `[1, 2, 3] == (1, 2, 3)`?

- A. True
- B. False
- C. TypeError
- D. SyntaxError

**Answer:** B

**Explanation:** List and tuple are different types, so they are not equal.

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Q22. What will this code print?

```
python
```

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```
def f(x, y=5): return x + y print(f(3))
```

- A. 8
- B. 3
- C. Error
- D. 5

**Answer:** A

**Explanation:** y uses default value 5.

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**Q23.** Which data type is unhashable (cannot be used as dict key)?

- A. int
- B. tuple
- C. list
- D. str

**Answer:** C

**Explanation:** Lists are mutable and unhashable.

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**Q24.** Which of the following is NOT true about Python modules?

- A. Modules are files containing Python code
- B. They can contain functions and classes
- C. Modules can be imported using import
- D. Modules are only for built-in functions

**Answer:** D

**Explanation:** Modules can be user-defined or built-in.

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**Q25.** What does `if __name__ == "__main__":` protect against?

- A. Syntax errors
- B. Code running on import
- C. Infinite loops
- D. Runtime exceptions

**Answer:** B

**Explanation:** Ensures code only runs when script executed directly.

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**Q26.** What happens when you assign a list to another variable like `b = a`?

- A. Creates a copy
- B. Points both to same list
- C. Error
- D. Deletes a

**Answer:** B

**Explanation:** Both refer to the same list object.

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**Q27.** What does the following code output?

```
python
```

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```
print(type(None))
```

- A. None
- B. <class 'NoneType'>
- C. null
- D. Error

**Answer:** B

**Explanation:** None is an object of type NoneType.

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**Q28.** Which of these correctly shows how to define a class in Python?

A.

```
python
```

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```
class Person: pass
```

B.

python

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```
class Person(): pass
```

C. Both A and B

D. Neither

**Answer: C**

**Explanation:** Both syntaxes are valid.

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**Q29.** Which of the following statements about indentation is true?

A. Tabs and spaces can be mixed freely

B. Consistent use of tabs or spaces is required

C. Indentation is optional

D. Indentation only affects comments

**Answer: B**

**Explanation:** Mixing tabs and spaces causes errors.

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Q30. What is the output of this code?

python

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```
def test(x): x += [4] return x print(test([1, 2, 3]))
```

- A. [1, 2, 3]
- B. [1, 2, 3, 4]
- C. Error
- D. [4]

**Answer:** B

**Explanation:** The list is mutated by appending 4.